
DAN KONDRATYUK

dankondratyuk@gmail.com ◇ dankondratyuk.com
linkedin.com/in/dankondratyuk ◇ github.com/hyperparticle

SUMMARY

ML Research Scientist and Co-Founder of [rekursiv.ai](#). Built large-scale multimodal systems spanning LLMs and diffusion models for joint image, video, audio, and text generation.

EXPERIENCE

rekursiv.ai 2026 - Present
Co-founder

- Building AI Scientist systems that hypothesize, design experiments, run them, interpret results, and discover new knowledge.
- Co-created open-source developer-agent infrastructure for typed, provider-portable coding agents and verified monorepo-to-OSS workflows.

Luma AI 2024 - 2025
Research Scientist; World Models Lead

- Co-developed the generative video models powering Dream Machine, Luma AI's core video generation product.
- Led the World Models team focused on realtime interactive GenAI video, including joint video-audio generation, video avatar speech-to-video, and realtime player and camera control.
- Contributed to large-scale multimodal foundation models for video, image, audio, and text generation.

Google Research 2019 - 2024
Senior Machine Learning Engineer; AI Resident *Seattle / Mountain View*

- First author and lead contributor of VideoPoet, a multimodal LLM for zero-shot video and audio generation; received an ICML 2024 Best Paper award.
- Created MoViNets, a family of efficient mobile video networks for realtime video action recognition on streaming video.
- Worked on Integrated Multimodal Perception, a mixture-of-experts model combining image, video, audio, and text modalities.
- Researched efficient training and inference techniques for video models, including model ensembling and cascades.

Hello Augi Oct 2016 - May 2019
Full Stack Developer *Boise*

- Created a contextually-aware dialog system for intelligent search of accumulated daily knowledge.
- Developed a Kotlin JVM backend on AWS integrating a Neo4j graph database with an Angular web client.

Sensus Aug 2015 - Oct 2016
Software Engineering Intern *Boise*

- Developed automated integration test libraries for a smart grid distributed system.
- Increased defect detection rate by 10% by implementing a Java radio simulator for SCADA hardware.

EDUCATION

- Charles University** — Prague, Czech Republic 2017 - 2019
M.S. Computational Linguistics; Best Master's Thesis Award, Mathematics & Physics department
- Saarland University** — Saarbrücken, Germany 2017 - 2019
M.S. Computational Linguistics; Erasmus Mundus Language & Communication Technologies
- Boise State University** — Boise, Idaho, USA 2013 - 2017
B.S. Computer Science, Mathematics Minor, Honors; Summa Cum Laude

SELECTED PUBLICATIONS

- Kondratyuk, D., Yu, L., Gu, X., Lezama, J., Huang, J., Hornung, R., Adam, H., Akbari, H., Alon, Y., Birodkar, V., et al. (2024). **VideoPoet: A Large Language Model for Zero-Shot Video Generation**. *ICML*. Best Paper.
- Kondratyuk, D., Yuan, L., Li, Y., Zhang, L., Tan, M., Brown, M., Gong, B. (2021). **MoViNets: Mobile Video Networks for Efficient Video Recognition**. *CVPR*.
- Kondratyuk, D., Straka, M. (2019). **75 Languages, 1 Model: Parsing Universal Dependencies Universally**. *EMNLP*.
- Akbari, H., Kondratyuk, D., Cui, Y., Hornung, R., Wang, H., Adam, H. (2023). **Alternating Gradient Descent and Mixture-of-Experts for Integrated Multimodal Perception**. *NeurIPS*.
- Yu, S., Hahn, M., Kondratyuk, D., Shin, J., Gupta, A., Lezama, J., Essa, I., Ross, D., et al. (2025). **MALT Diffusion: Memory-Augmented Latent Transformers for Any-Length Video Generation**. *arXiv*.
- Marmon, A., Schindler, G., Lezama, J., Kondratyuk, D., Seybold, B., Essa, I. (2024). **CamViG: Camera Aware Image-to-Video Generation with Multimodal Transformers**. *arXiv*.
- Wang, X., Kondratyuk, D., Kitani, K., Movshovitz-Attias, Y., Eban, E. (2021). **Wisdom of Committees: An Overlooked Approach To Faster and More Accurate Models**. *arXiv*.
- Kondratyuk, D., Tan, M., Brown, M., Gong, B. (2020). **When Ensembling Smaller Models is More Efficient than Single Large Models**. *WebVision CVPR Workshop*.
- Kondratyuk, D. (2019). **Cross-Lingual Lemmatization and Morphology Tagging with Two-Stage Multilingual BERT Fine-Tuning**. *SIGMORPHON*.
- Kondratyuk, D., Gavenčiak, T., Straka, M., Hajič, J. (2018). **LemmaTag: Jointly Tagging and Lemmatizing for Morphologically Rich Languages with BRNNs**. *EMNLP*.
- Kondratyuk, D., Kennington C. (2017). **Towards a Dialogue System with Long-term, Episodic Memory**. *SEMDIAL*.

HONORS AND AWARDS

- **ICML Best Paper** (2024). *VideoPoet: A Large Language Model for Zero-Shot Video Generation*.
- **Google AI Residency** (2019 - 2021). *Google Research*. Industry research program.
- **Best Master's Thesis Award** (2019). *Charles University in Prague*. Awarded by the Mathematics & Physics department for multilingual dependency parsing research.
- **1st Place in Morphology Tagging** (2019). *SIGMORPHON Shared Task #2*.
- **Erasmus Mundus Joint Master's Degree Scholarship** (2017 - 2019). *Language & Communication Technologies*. European Commission scholarship covering tuition, travel, and living expenses.
- **Best Poster Presentation Award** (2019). *Language & Communication Technologies*.
- **Outstanding Student Award** (2017). *Boise State University Computer Science Department*.
- **Distinguished Honors Scholar** (2017). *Boise State University Honors College*.
- **REU in Software Security** (2015). *Boise State University*. NSF Grant No. CNS 1461133.

-
- **REU in Mathematics** (2015). *Boise State University*. NSF Grant No. DMS 1062857.

SELECTED PROJECTS

rekursiv.ai

2026 - Present

Co-founder

- Building AI Scientist systems that autonomously create knowledge through hypothesis generation, experiment design, execution, and interpretation.
- Open-source work includes *sagent*, a typed Python API and CLI for coding agents, and *copybarista*, tooling for safe OSS exports from private monorepos.

Dream Machine

2024 - 2025

Luma AI

- Co-developed production generative video models for Luma AI's Dream Machine product.
- Led realtime interactive video model work spanning video-audio generation, speech-to-video avatars, and player/camera control.

VideoPoet

2023 - 2024

Google Research

- Led development of a multimodal LLM for zero-shot video and audio generation from images, video, text, and audio conditioning signals.
- Project page: <https://sites.research.google/videopoet/>

Mobile Video Networks (MoViNets)

2019 - 2021

Google Research

- Created efficient streaming 3D CNNs for mobile video action recognition, reducing memory usage from linear to constant in video duration.
- Project: <https://arxiv.org/abs/2103.11511>

UDify: Multilingual Universal Dependency Parser

2018 - 2019

Charles University & Saarland University

- Designed a single multilingual model for universal POS tags, morphology, lemmas, and dependency trees across 75 languages.
- Open source: <https://github.com/Hyperparticle/udify>